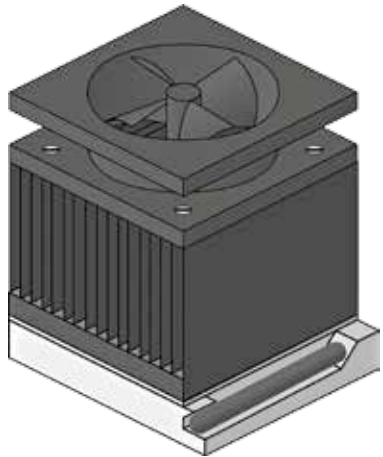


these two extremes lie a plethora of cooling solutions for different needs and for different components. Based on what you want to cool and how much you wish to cool it down by, you have even more options. Let's start with your basic CPU cooler as an example.

Basic anatomy of a cooler

It's a fact that coolers operate on the principles of physics and for that reason, the basic design of coolers are more or less the same. At the base lies the mounting mechanism. Different CPUs and GPUs have standards for mounting coolers onto them based on the TDP range of the CPUs in use. Your average mainstream Intel Core i3/i5/i7 and AMD Ryzen 3/5/7 CPUs tend to start at about 40 W TDP and peak at 105 W TDP. Sometimes, the peak limit can be further up. The sockets for these mainstream CPUs are practically the same for a particular company. Intel has the LGA1151 socket for mainstream CPUs right now which supports mainstream Core i3, i5 and i7 SKUs with up to 8-cores. On AMD's side you have the AM4 socket which supports the first and second Ryzen generation of CPUs. However, both CPU manufacturers have HEDT (High-End Desktop) CPUs which use different sockets. Intel's HEDT platform uses the LGA2066 socket and AMD has the TR4 socket for its HEDT platform. Usually, cooler manufacturers make coolers that support all sockets and have adapters to make them work for different sockets. Using a normal cooler for an HEDT CPU is not advisable since the TDP value for the CPU will be higher than what normal coolers are designed to dissipate.



The average CPU cooler.

That being said, your average cooler is made up of a couple of parts. With passive coolers, you'll have just the mounting plate or screws and a reasonably large heatsink. That's it. With active coolers, you will have a mounting mechanism, a heatsink that may be made of more than one metal and then you'll have a cooling mechanism.